📊 Slide 1 — Data Acquisition & Quality

Source, Period, Quality Issues & Resolutions

DATA SOURCE & PERIOD

- Provider: Yahoo Finance (yfinance Python library)
- Period: 01 Jan 2023 – 31 Dec 2024 (2 full calendar years)
- Frequency: Daily OHLCV (Adjusted Close for all return calculations)
- Universe: 15 NSE-listed stocks across 3 sectors

  Banking (5): HDFCBANK, ICICIBANK, SBIN, KOTAKBANK, AXISBANK

  IT (5): TCS, INFY, WIPRO, HCLTECH, TECHM

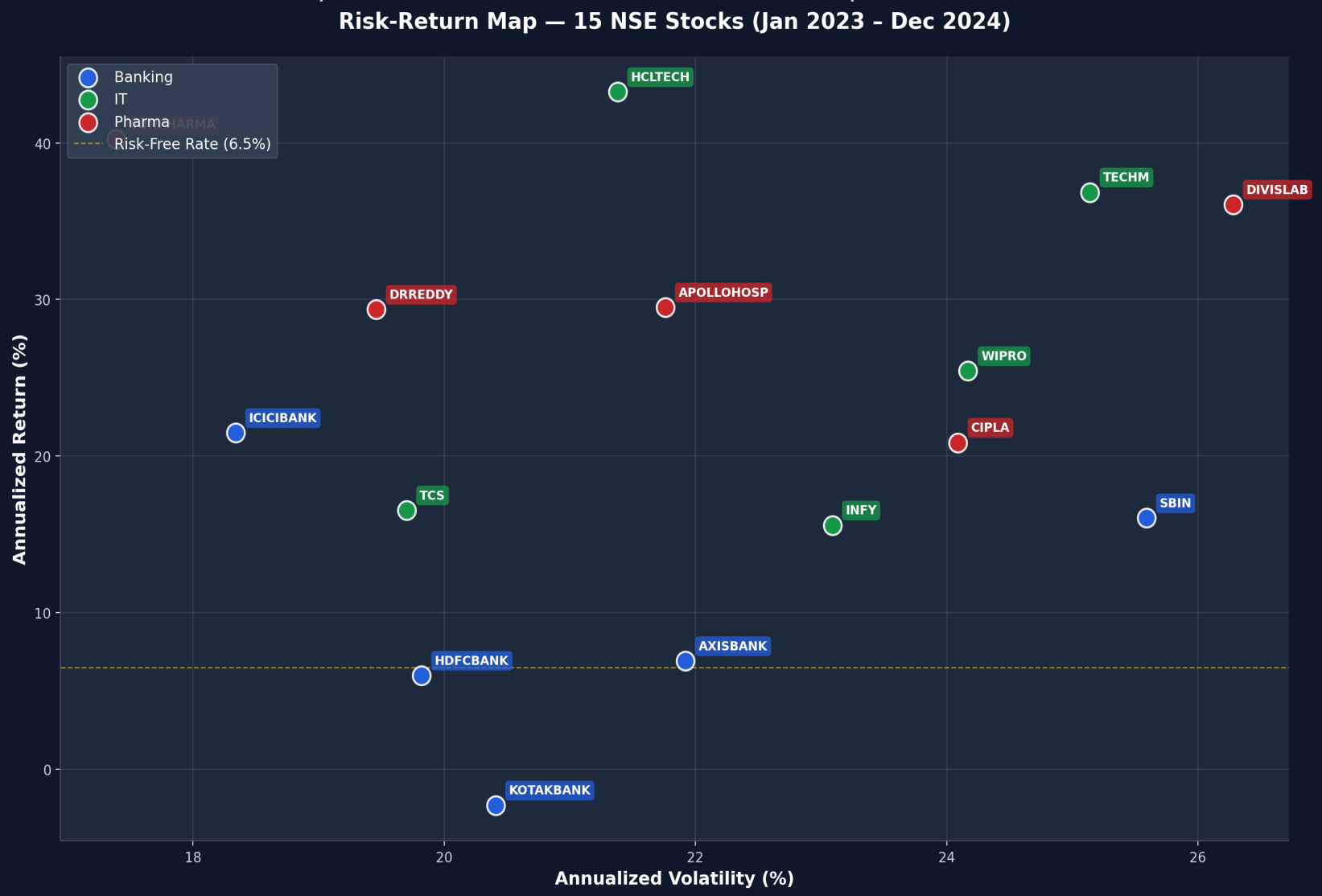
  Pharma (5): SUNPHARMA, DRREDDY, CIPLA, DIVISLAB, APOLLOHOSP
- Benchmark: Nifty 50 Index (^NSEI)

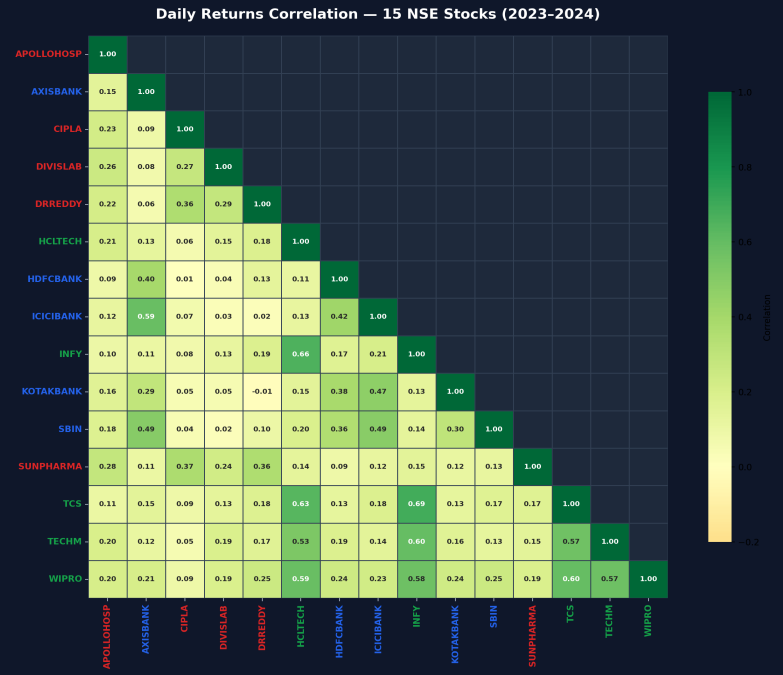
DATA QUALITY ACTIONS

- Missing values → Forward-filled, then back-filled
- Indian market holidays → Expected gaps, no additional treatment
- Zero/negative prices → Replaced with forward-fill
- Extreme returns (>25% daily) → Flagged but retained (legitimate events)
- Validated: No stock has >5 consecutive missing trading days
- All timestamps timezone-normalized (IST assumed)

📈 Slide 2 — Risk-Return Map

Annualized Volatility vs Annualized Return — Sector Clusters Explained





Daily Returns Correlation Heatmap

KEY OBSERVATIONS

- Banking cluster: Moderate vol, correlated
- IT cluster: Varies widely in risk-return
- Pharma: Defensive, lower Beta
- Risk-free rate line at 6.5% p.a.

🏆 Slide 3 — Top & Bottom Stocks by Sharpe Ratio

Best 3 and Worst 3 performers — What separates winners from laggards

🟢 TOP 3 (Highest Sharpe Ratio)

Ticker	Sector	Annualized Return (%)	Annualized Volatility (%)	Sharpe Ratio	Beta vs Nifty 50	Maximum Drawdown (%)
SUNPHARMA	Pharma	40.24	17.39	1.9399	0.4725	-12.24
HCLTECH	IT	43.26	21.38	1.719	0.7871	-21.66
TECHM	IT	36.85	25.14	1.2074	0.8729	-16.22

🔴 BOTTOM 3 (Lowest Sharpe Ratio)

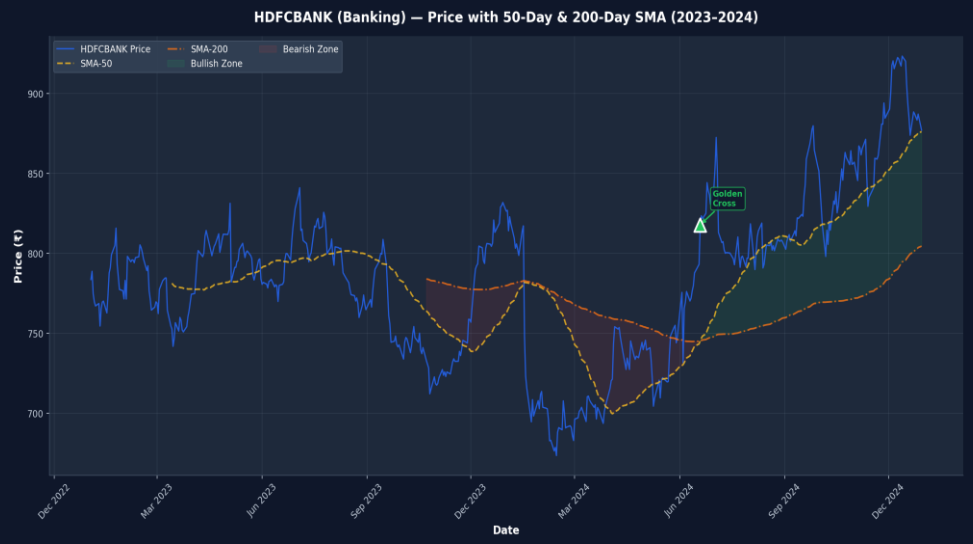
Ticker	Sector	Annualized Return (%)	Annualized Volatility (%)	Sharpe Ratio	Beta vs Nifty 50	Maximum Drawdown (%)
AXISBANK	Banking	6.91	21.92	0.0185	1.0601	-18.78
HDFCBANK	Banking	5.98	19.82	-0.0265	1.0391	-19.91
KOTAKBANK	Banking	-2.31	20.41	-0.4318	0.8634	-23.15

The top 3 stocks are separated by stronger risk-adjusted performance: they generated high annualised returns without taking proportionally high volatility, which gave them better Sharpe ratios. SUNPHARMA and DRREDDY also benefited from lower beta, so they were less sensitive to market swings. HCLTECH stood out because its return was very strong even after accounting for risk. The bottom 3 are separated by weaker reward for the risk taken: they either delivered low or negative returns, or had high volatility and drawdowns that were not justified by performance. In short, the best stocks were more efficient, while the worst stocks took risk without enough payoff.

📉 Slide 4 — SMA Technical Signals

50-Day & 200-Day SMA — Golden Cross / Death Cross as of 31 Dec 2024

Ticker	Sector	SMA-50	SMA-200	Signal	Last Crossover Date
APOLLOHOSP	Pharma	7050.71	6559.55	Golden Cross	nan
AXISBANK	Banking	1144.55	1160.8	Death Cross	2024-12-12
CIPLA	Pharma	1487.43	1492.87	Death Cross	2024-12-24
DIVISLAB	Pharma	5901.81	4838.6	Golden Cross	2024-05-16
DRREDDY	Pharma	1263.89	1267.88	Death Cross	2024-12-13
HCLTECH	IT	1792.28	1558.68	Golden Cross	2024-07-22
HDFCBANK	Banking	876.15	804.55	Golden Cross	2024-06-19
ICICIBANK	Banking	1282.17	1188.17	Golden Cross	nan
INFY	IT	1825.72	1648.63	Golden Cross	2024-07-12
KOTAKBANK	Banking	352.1	353.36	Death Cross	2024-12-26
SBIN	Banking	812.27	795.13	Golden Cross	nan
SUNPHARMA	Pharma	1789.64	1661.11	Golden Cross	nan
TCS	IT	3988.23	3898.45	Golden Cross	nan
TECHM	IT	1665.61	1439.69	Golden Cross	nan
WIPRO	IT	271.8	244.06	Golden Cross	nan



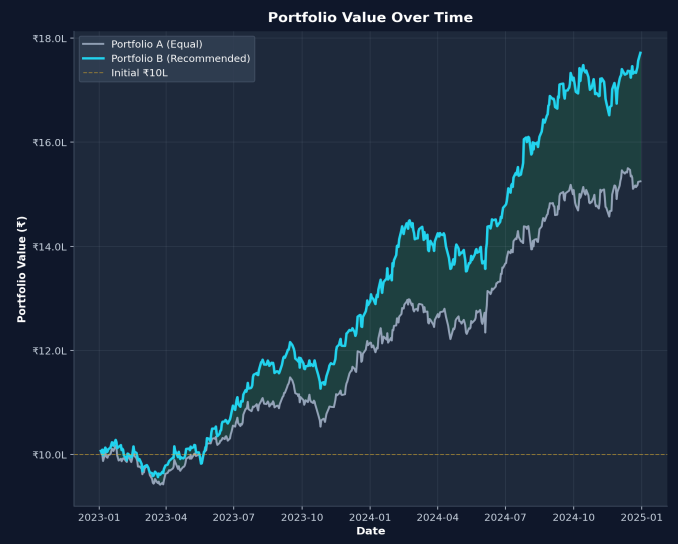
HDFCBANK — SMA Overlay (Banking)

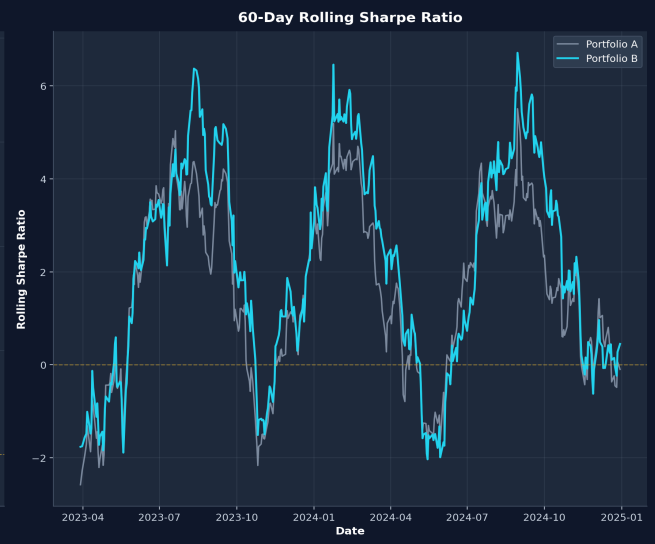
TCS — SMA Overlay (IT)

💼 Slide 5 — Portfolio Recommendation + Chaos Round

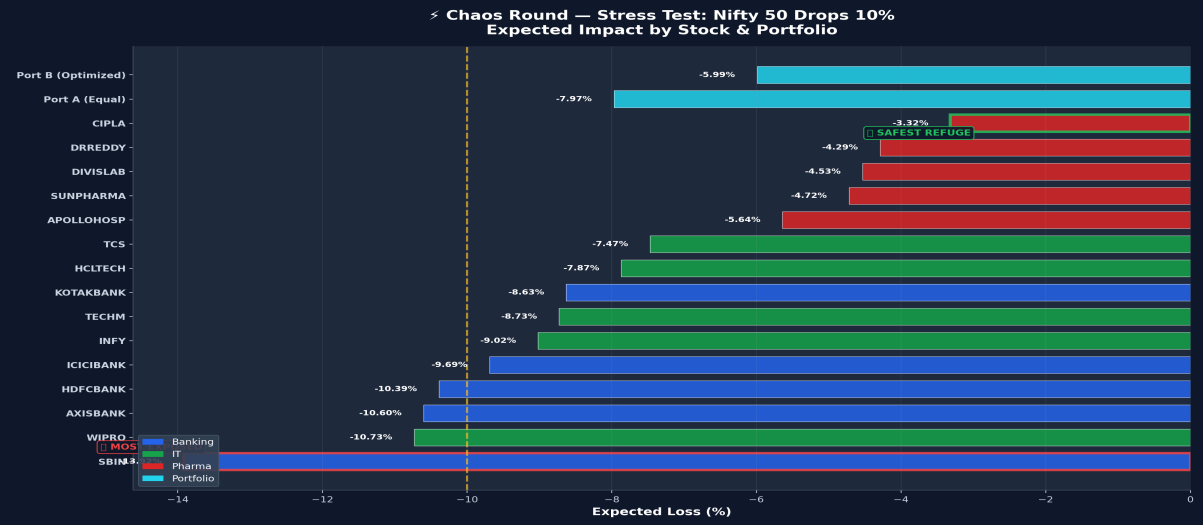
Portfolio A (Equal) vs Portfolio B (Optimized) + Nifty -10% Stress Test

Metric	Portfolio A (Equal Weight)	Portfolio B (Recommended)
Annualized Return (%)	24.28	34.29
Annualized Volatility (%)	11.43	12.1
Sharpe Ratio	1.5564	2.2956
Banking Exposure (%)	33.3	5.7
IT Exposure (%)	33.3	27.7
Pharma Exposure (%)	33.3	66.6





⚡ Chaos Round: Nifty -10% Stress Test



Ticker	Sector	Beta	Expected Loss (%)
PORTFOLIO A	Portfolio	-	-7.97
PORTFOLIO B	Portfolio	-	-5.99
MOST EXPOSED	Banking	1.3916	-13.92
SAFEST REFUGE	Pharma	0.3322	-3.32

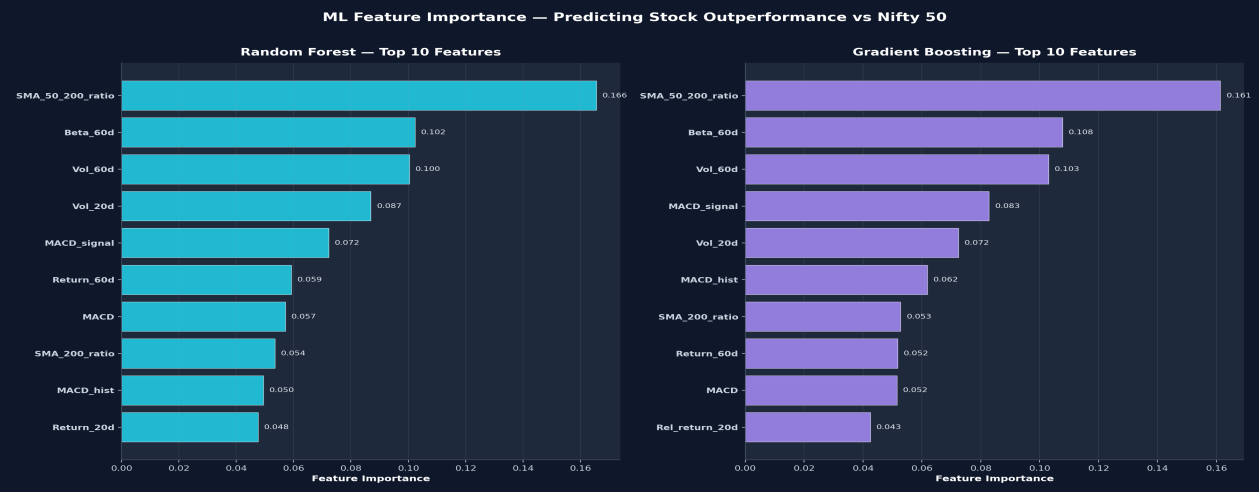
Portfolio B is strategically allocated to minimize drawdown using a defensive pivot.

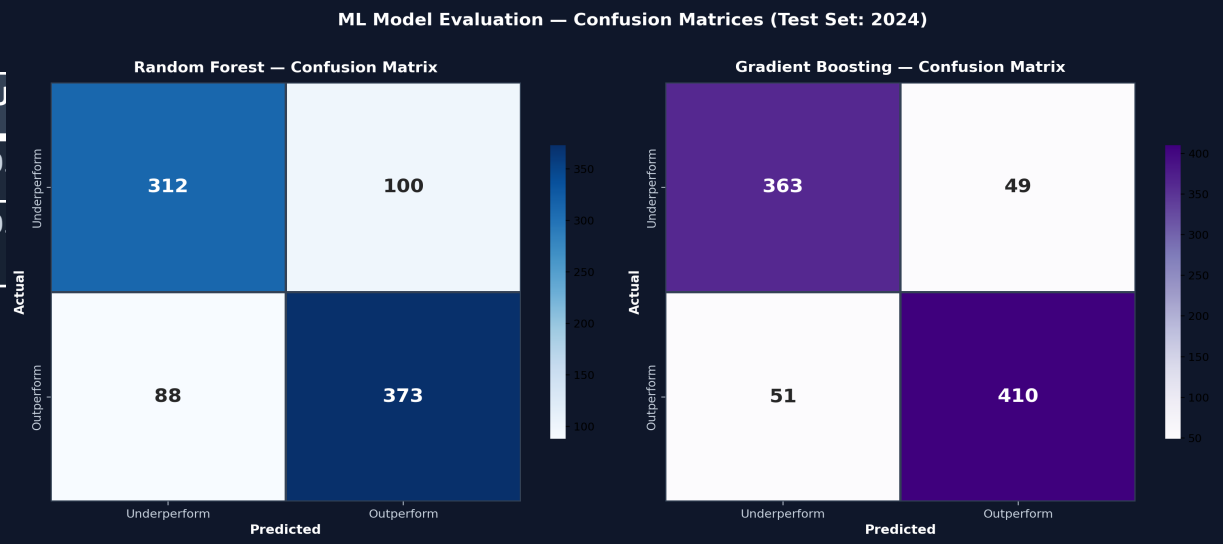
🤖 Slide 6 — ML Model Comparison

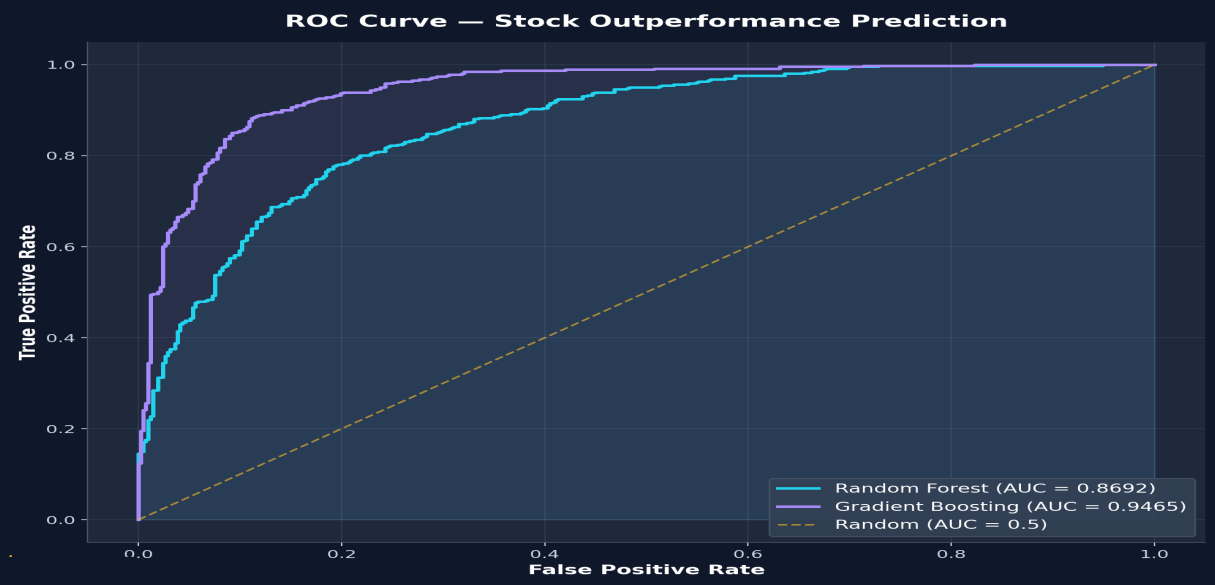
Random Forest vs Gradient Boosting — Predicting Stock Outperformance vs Nifty 50 (20-day forward)

Model Performance (Test: 2024)

Model	Accuracy	Precision	Recall	F1 Score	AU
Random Forest	0.7847	0.7886	0.8091	0.7987	0.
Gradient Boosting	0.8855	0.8932	0.8894	0.8913	0.





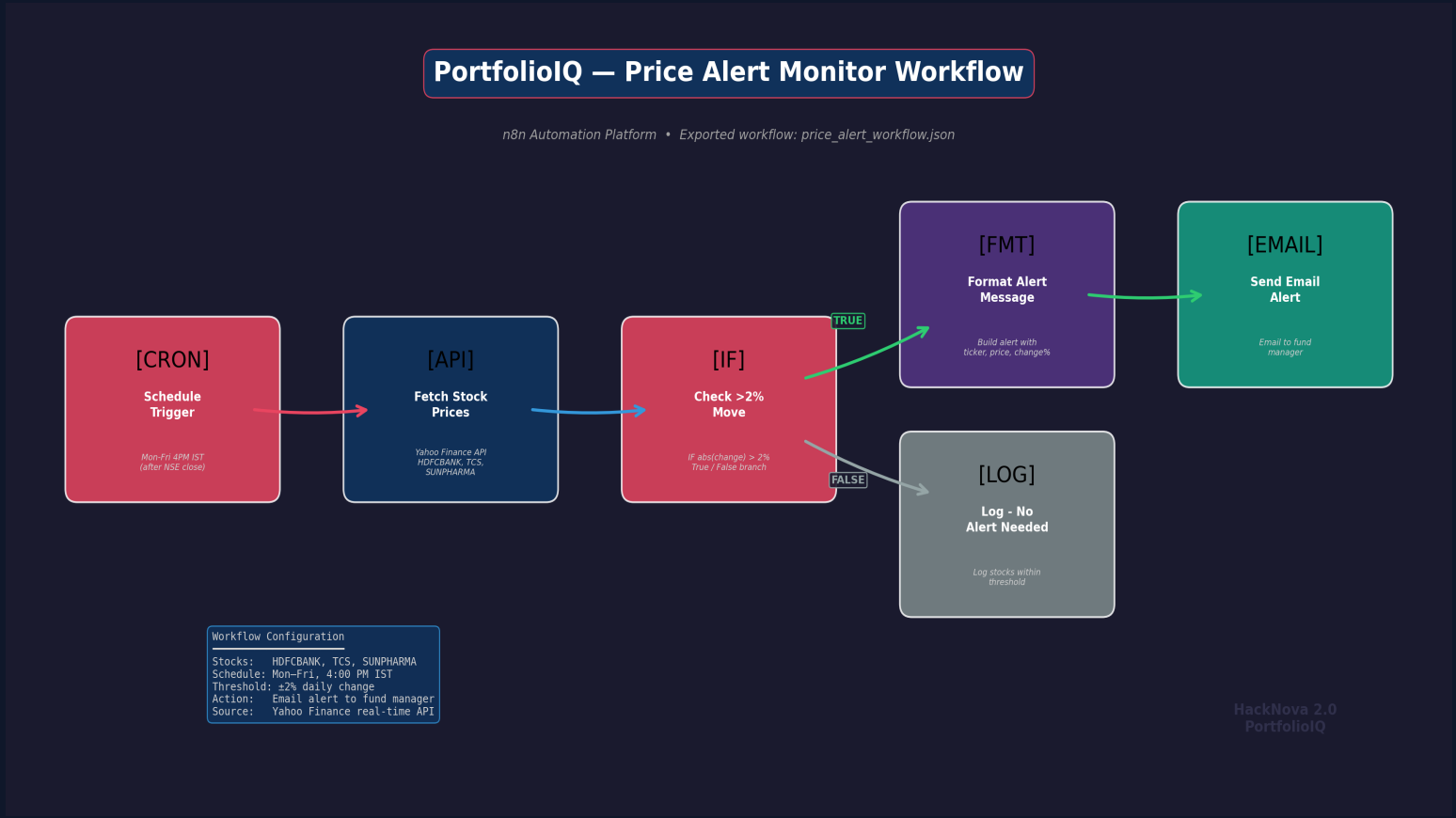


ML Design:

- Target: Does stock outperform Nifty 50 over next 20 days?
- Features: 18 technical indicators (returns, volatility, SMA ratios, RSI, MACD, Beta, volume)
- Temporal split: Train on 2023, Test on 2024 (no data leakage)
- Class balancing: balanced weights (RF) / scale_pos_weight (XGB)

⚙️ Slide 7 — n8n Automation Demo

Price Alert Workflow — Live Execution



WORKFLOW: PRICE ALERT MONITOR

Purpose:

Automated monitoring replaces manual daily checks, ensuring the fund manager is immediately notified of significant price movements requiring action.

Configuration:

- Stocks: HDFCBANK, TCS, SUNPHARMA
- Schedule: Mon–Fri at 4:00 PM IST (post NSE close)
- Threshold: Alert if any stock moves >2%
- Action: Email alert with ticker, price, change %

📺 **LIVE DEMO:**

Import JSON → Execute workflow → Show alert output

Workflow JSON → n8n/price_alert_workflow.json | Runs Mon-Fri 4 PM IST | Uses Yahoo Finance API | Sends alert via email to fund manager